



Figure 6-1: The set $\{A, B, C\}$ within the dashed lines and some of its subsets $\{A, B, C\}$, $\{A, C\}$, and $\{\}$ within the solid lines. The circles represent sets, and the letters represent elements.

A *permutation* of a set is a specific ordering of all elements in the set. For example, the set $\{A, B, C\}$ has six permutations: ABC, ACB, BAC, BCA, CAB, and CBA. We call these *permutations without repetition*, or *permutations without replacement*, because each element doesn't appear in the permutation more than once.

A *combination* is a selection of elements of a set. More formally, a *k-combination* is a subset of k elements from a set. Unlike permutations, combinations don't have an ordering. For example, the 2-combinations of the set $\{A, B, C\}$ are $\{A, B\}$, $\{A, C\}$, and $\{B, C\}$. The 3-combination of the set $\{A, B, C\}$ is $\{A, B, C\}$.

The term *n choose k* refers to the number of possible combinations (without repetition) of k elements that can be selected from a set of n elements. (Some mathematicians use the term *n choose r*.) This concept has nothing to do with the elements themselves, just the number of them. For example, 4 choose 2 is 6, because there are six ways to choose two elements from a set of four elements like $\{A, B, C, D\}$: $\{A, B\}$, $\{A, C\}$, $\{A, D\}$, $\{B, C\}$, $\{B, D\}$, and $\{C, D\}$. Meanwhile, 3 choose 3 is 1, because there's only one 3-combination from a set of three elements like $\{A, B, C\}$; that is, $\{A, B, C\}$ itself. The formula for calculating *n choose k* is $(n!) / (k! \times (n - k)!)$. Recall that $n!$ is the notation for factorials: $5!$ is $5 \times 4 \times 3 \times 2 \times 1$.

The term *n multichoose k* refers to the number of possible combinations *with repetition* of k elements that can be selected from a set of n elements. Because k -combinations are sets and sets do not have duplicate elements, a k -combination does not have repetition. When we use k -combinations with